

PARTS LIST

Semi-conductors:

- IC1 - CD4060 oscillator-cum-divider
 IC2 - 555 timer
 T1 - BC557 pnp transistor
 T2 - BC547 npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 100-kilo-ohm
 R2 - 120-ohm
 R3, R5 - 4.7-kilo-ohm
 R4 - 22-kilo-ohm
 R6 - 1-kilo-ohm
 R7 - 56-kilo-ohm

Capacitors:

- C1 - 1nF ceramic disk
 C2, C3 - 0.1 μ F ceramic disk
 C4, C8 - 0.01 μ F ceramic disk
 C6 - 100 μ F, 25V electrolytic
 C5 - 2.2 μ F, 25V electrolytic
 C7 - 0.047 μ F ceramic disk
 C9 - 47 μ F, 25V electrolytic

Miscellaneous:

- L1 - 100 μ H inductor
 P1 - Touch plate
 LS1 - 8-ohm loudspeaker
 CON1 - 2-pin connector for battery connection
 - 12V adaptor

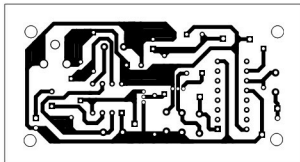


Fig. 2: An actual-size, single-side PCB for the touch alarm

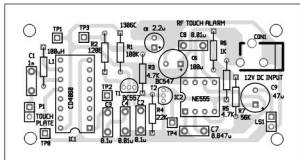


Fig. 3: Component layout for the PCB

is non-conducting, pin 4 of IC2 is high, activating the alarm. The frequency of sound can be changed by changing

the value of timing components R6, R7 and C7.

Construction and testing

An actual-size, single-side PCB for the touch alarm is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on a PCB, enclose it in a suitable case.

To test the circuit, check correct power supply at TP1 with respect to TP0. Do not touch the touch plate and verify 1kHz (approx.) frequency at TP2. This frequency will die if you touch the touch plate. At TP3 and TP4, verify voltage levels as indicated in the table. ●

The author is a regular contributor to EFY

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MIRACLE ELECTRONIC DEVICES PVT. LTD.

Regd. Office : # 48/H, Korihobanahalli Village, Thigalarapalya Main Road,
 Peenya 2nd Stage, Bangalore - 560 059, (India)
 Tel: +91 80 28362750/ 51 | Fax: +91 80 2836 2752
 sales@miracle.net.in | www.toroidel.com